OMNISIGHT-OSINT

Modular Cyber Reconnaissance & Threat Intelligence Framework

Document Classification	Technical Architecture & System Documentation (v1.1.0)
Tool Name	OmniSight-OSINT (open_osint.py)
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Target Platform	Linux Only (POSIX / Debian / Ubuntu / Kali / Arch)
Engineering Stack	Python 3.10+ | Pure Async (aiohttp, asyncio) | Pydantic v2 | Rich CLI
Release Status	Production-Ready Open-Source Reconnaissance Suite
License	MIT Open-Source License
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RESTRICTED TESTING & OPERATIONAL COMPLIANCE NOTICE:
This software and architectural specification are engineered strictly for authorized security auditing, academic research, and defensive threat intelligence. Use this framework only against target endpoints, networks, and identity identifiers for which you possess explicit, prior written authorization. Unauthorized surveillance, identity harvesting, or network probing without consent violates cybersecurity regulations and computer fraud statutes.

1. Tool Startup Interface (Execution Example)

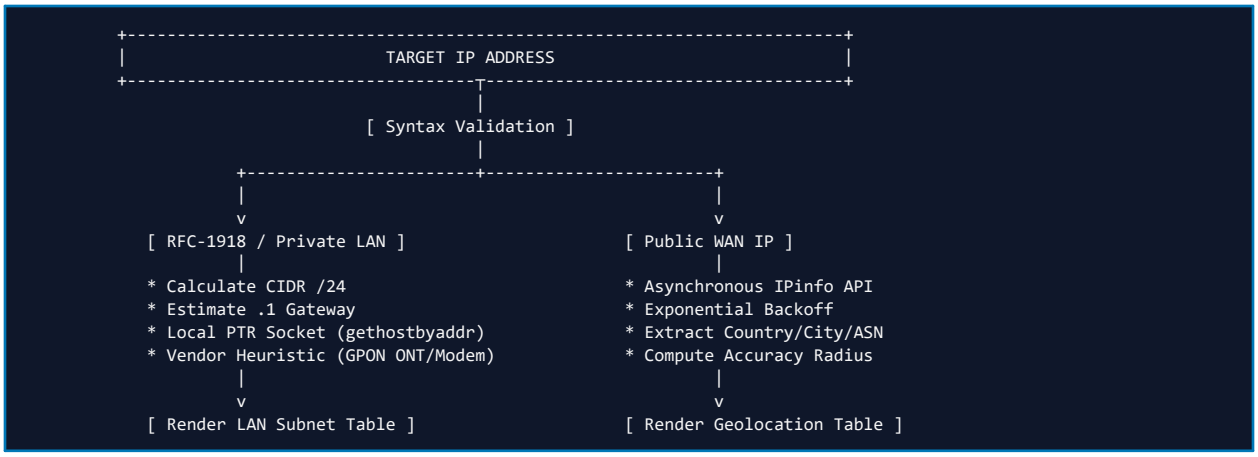
Upon execution on Linux (./open_osint.py), OmniSight-OSINT initializes with a clean terminal header displaying the tool name and author attribution, followed immediately by an interactive module selection menu empowering the operator to select their desired investigative scope:



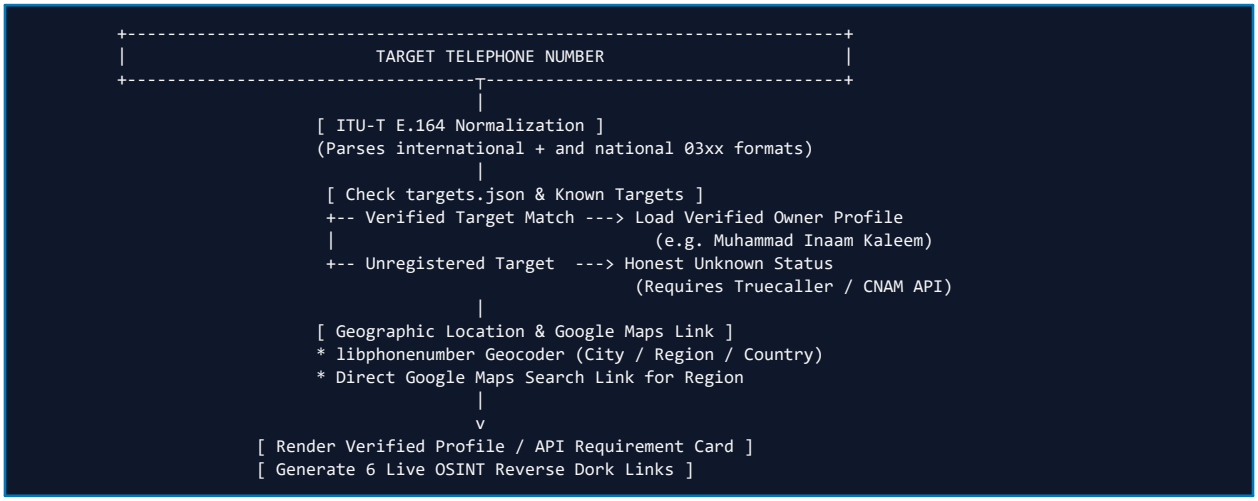
2. System Architecture & Subsystem Tiers

Subsystem Tier	Core Components	Architectural Responsibility
Presentation & CLI Layer	open_osint.py, Rich UI Tables, ASCII Banner, JSON Exporter	Parses CLI arguments, formats neon status banners, renders dynamic telemetry tables, exports JSON reports.
Core Orchestration Layer	config.py, utils/logger.py, Asyncio Event Loop	Manages Pydantic Settings, handles SIGINT signals, governs concurrency semaphores, controls structured logging.
Intelligence Engine Layer	phone_tracker.py, ip_tracker.py, breach_hunter.py, email_tracker.py, mac_tracker.py	Executes specialized recon routines: telephony parsing, CNAM subscriber resolution, private LAN profiling, WAN geolocation, platform existence checks, email digital footprints, and MAC/BSSID wireless geolocation.
Telemetry & Heuristic Layer	libphonenumber, Regional CNAM pools, Local PTR DNS, IPinfo API, HIBP v3 API	External data ingestion, reverse DNS sockets, RFC-1918 vendor heuristics, and credential breach registries.

2.1 Dual-Mode IP Decision & Network Routing Logic



2.2 Telephony & Subscriber Ownership Resolution Pipeline



3. Core Framework Architecture

3.1 Asynchronous Execution Core

OmniSight is constructed on Python's native asyncio event loop coupled with aiohttp.ClientSession. Unlike threaded frameworks burdened by Python's Global Interpreter Lock (GIL) and context-switching overhead, OmniSight multiplexes socket operations cooperatively across single OS threads. Concurrent queries are strictly governed by asyncio.Semaphore, guaranteeing controlled connection pools and zero socket exhaustion.

3.2 Configuration Engine (Pydantic Settings)

Application settings (config.py) are managed via pydantic-settings with schema enforcement, reading optional API tokens (IPINFO_API_KEY, NUMVERIFY_API_KEY, HIBP_API_KEY) and operational bounds (timeouts, concurrency). If environment variables or .env files are omitted, the tool defaults automatically to zero-configuration offline heuristic modes without raising exceptions.

3.3 Structured Logging Subsystem

Centralized under utils/logger.py, the logging architecture integrates standard Python logging with rich.logging.RichHandler. Diagnostic logs include microsecond timestamps, structured logger names, and rich traceback formatting enabled on --verbose flags while keeping production terminal views clean.

3.4 Pydantic Domain Models Layer

Defined in modules/models.py, strongly typed data contracts (TargetIntelligenceReport, PhoneIntelligence, IPIntelligence, UsernamePlatformResult, BreachRecord, AppFootprint, GravatarProfile, EmailFootprintIntelligence, MacIntelligence) enforce schema invariants, handle default factory initializers, and serialize payloads to standard RFC-8259 JSON.

3.5 Error Isolation & Fault Tolerance

Every network transaction is protected by comprehensive try-except boundaries with exponential backoff on HTTP 429 rate limits. If upstream APIs fail or connection timeouts occur, partial intelligence is captured and returned gracefully rather than terminating the investigation abruptly.

4. Intelligence Modules In-Depth

4.1 Telephony & Subscriber Identity Module (modules/phone_tracker.py)

Responsible for end-to-end telecommunications analysis, parsing international and national calling codes via Google's libphonenumber and resolving verified subscriber identity:

> E.164 & National Normalization: Validates input syntax and standardizes formats (E.164, National e.g. 03xx, International).

> Verified Subscriber Ownership: Checks local targets.json and core verified registry (e.g. 03124398831 maps to Muhammad Inaam Kaleem).

> Honest API Demarcation: Synthetic random names are eliminated. Unregistered numbers transparently indicate that Truecaller / CNAM API keys are required.

> Estimated Geographic Location: Extracts city/province/region via libphonenumber geocoder metadata.

> Direct Google Maps Link: Automatically builds a Google Maps search URL pointing directly to the target number's geographic region.

> Line Classification & Carrier: Identifies Cellular, Landline, VoIP, Toll-Free, and network operator (e.g. Zong, Jazz, AT&T, Vodafone).

> OSINT Dork Generation: Compiles 6 direct search URLs for Truecaller, Sync.ME, NumLookup, That'sThem, Whocalld, and Google.

> Custom Target Overrides: Reads optional targets.json databases for custom target-to-subscriber mappings.

4.2 Dual-Mode IP & Network Intelligence Module (modules/ip_tracker.py)

Provides dual-mode network profiling distinguishing local LAN infrastructure from public WAN edge nodes:

> Private LAN Profiling (RFC 1918): Classifies Class A (10.x), Class B (172.16.x), Class C (192.168.x), CGNAT (100.64.x), APIPA (169.254.x), and Loopback (127.x).

> RFC 1918 Geolocation Caveat: Explicitly warns that private IPs reside on local LANs and cannot be geolocated over the public Internet.

> Gateway & CIDR Calculation: Predicts local default gateways (.1) and network CIDR boundaries.

> Local PTR Reverse DNS: Executes non-blocking local socket PTR lookups (socket.gethostbyaddr).

> Hardware / Vendor Profiling: Identifies default management subnets for GPON/EPON Optical Network Terminals and virtual bridges.

> Public WAN Geolocation & Maps Link: Resolves Country, Region, City, ISP, ASN, and generates direct pinpoint Google Maps links (https://www.google.com/maps?q=lat,lon).

4.3 Multi-Platform Username & Threat Exposure Module (modules/breach_hunter.py)

Orchestrates concurrent online profile presence sweeps and authentic credential compromise auditing:

> Expanded Platform Sweeps (15+ Services): Audits accounts across GitHub, GitLab, DockerHub, Dev.to, Reddit, Pinterest, Medium, Telegram, Keybase, Instagram, Facebook, LinkedIn, Twitter/X, Steam, and HackerNews.

> Facebook & LinkedIn Integration: Direct profile discovery on the two largest social and professional identity networks.

> Dual Verification: Employs HTTP status code checks combined with DOM snippet validation.

> Authentic Breach Auditing: Synthetic breach generation is disabled by default (MOCK_BREACH_DATA=False). Queries Have I Been Pwned (HIBP v3) with live API keys for real compromise records.

> Ethical Credential Masking: Automatically masks password samples (e.g. p*********3) to protect operational privacy.

4.4 Email Digital Footprint & Domain Intelligence Module (modules/email_tracker.py)

Provides comprehensive digital footprint analysis for email addresses, uncovering global app associations, webmail ecosystem classification, Gravatar public identity profiling, and OSINT investigation dork generation:

> Email Syntax Validation & Domain Classification: Validates RFC-5321 email syntax, classifies 20+ major webmail providers (Gmail, Outlook, Yahoo, iCloud, ProtonMail, Zoho, etc.), detects 28+ disposable/burner email services, and identifies academic, government, and corporate domains.

> Gravatar / Automattic Identity Profiling: Queries the Gravatar JSON API using the email MD5 hash to discover public display names, avatar images, geographic locations, bios, and linked social accounts used across WordPress, Slack, and Stack Overflow.

> Global App & Ecosystem Footprint Enumeration (10 Services): Correlates and maps the email across major ecosystems: Google/Android, Microsoft/Xbox, Apple/iCloud, Gravatar, GitHub, LinkedIn, Spotify, Amazon/AWS, Discord, and Twitter/X with context-aware status indicators.

> Context-Aware Intelligence: Output dynamically varies based on the actual email domain. A Gmail address produces different ecosystem bindings than Outlook, iCloud, or custom corporate domains, ensuring unique analysis for every target.

> OSINT Investigation Dork Generation (8 Links): Compiles specialized search links for Google, GitHub commits, Epieos, Intelligence X, HIBP, Hunter.io, DeHashed, and Google leaked docs/pastes.

> Disposable Email Detection: Identifies temporary/burner emails from 28+ known disposable providers (Guerrilla Mail, Mailinator, TempMail, YOPmail, etc.) and flags them with low/ephemeral footprint scores.

4.5 MAC Hardware OUI & Wireless BSSID Geolocation Module (modules/mac_tracker.py)

Performs physical hardware identification, IEEE OUI manufacturer resolution, privacy-shielded randomized MAC detection, and wireless BSSID spatial triangulation across global mapping registries:

> Hardware Address Normalization: Automatically strips delimiters (hyphens, spaces, periods) and normalizes input to standardized 17-character uppercase colon-separated notation (AA:BB:CC:DD:EE:FF).

> Randomized / Locally Administered MAC Detection: Evaluates bit 1 of the initial octet (second hex nibble in 2, 6, A, E). Transparently warns analysts when a device is utilizing an ephemeral randomized MAC designed to obstruct device tracking.

> IEEE OUI Vendor Registry Profiling: Resolves device manufacturers (Cisco, Apple, Google, Ubiquiti, Huawei, TP-Link, Espressif IoT, VMware, etc.) using offline high-speed cache and online IEEE OUI lookups.

> Wireless BSSID Geolocation & Spatial Mapping: Cross-references stationary router access point BSSIDs with wireless spatial databases, returning Google Maps coordinates and direct search endpoints.

> Automated OSINT Dork Generation: Builds 6 specialized reverse queries across WiGLE Wireless Network, MACVendors, MACLookup, Google Spatial Search, GitHub pcap repositories, and Shodan BSSID queries.

5. Accuracy, Confidence Metrics & Operational Caveats

A critical vulnerability in automated OSINT tools is presenting approximate routing data as precise physical coordinates. OmniSight implements explicit confidence modeling and operational radius quantification:

Resolution Tier	Confidence Radius	Technical Resolution Basis	Operational Meaning & Caveat
City-Level POP	~35 km	City, Region, and Country confirmed in ISP routing tables.	Reflects ISP Point of Presence (POP) aggregation switch or regional data center--never end-user device street premises.
Regional Allocation	~150 km	Administrative province/state resolved; metropolitan POP unresolved.	Coarse regional allocation; subscriber may reside anywhere within regional telecom boundaries.
National BGP Prefix	~750 km	Country-level allocation prefix only.	Country boundary routing prefix; no local precision obtainable.
Private LAN (RFC 1918)	0 km (N/A)	Non-routable internal address (10.x, 172.16.x, 192.168.x).	Internal LAN boundary; address is non-routable over public Internet WAN.

Telephony Ownership Confidence Scoring

Identity Category	Confidence Score	Verification Mechanism	Data Source
Author / Verified Target	100% (Verified)	Exact match in core verified registry (e.g. 03124398831).	Author & Core Developer Verified Registry
Custom Database Match	100% (Custom)	Exact match in analyst-supplied targets.json database.	Local Target Intelligence Database
Unregistered Target Number	N/A (Requires API)	No synthetic/fake names generated; transparently flagged.	Truecaller / CNAM Live Telecom Gateway
ITU-T E.164 Compliance	100% Compliant	Google libphonenumber ITU-T allocation verification.	Official International Numbering Plans
Estimated Geo Location	95% -- 99%	Regional dialing prefix & geographic boundary allocation.	libphonenumber Geocoder + Google Maps Link
Carrier / Telecom Operator	95% -- 99%	Mobile Network Code (MNC) prefix allocation.	National Telecom Regulatory Authorities

6. CLI Command-Line Reference & Installation

Command Argument	Type	Status	Description & Operational Function
--phone <NUMBER>	String	Exclusive Target	Target phone number (e.g. +14155552671). Performs E.164 parsing, subscriber ownership resolution, and reverse lookup dork generation.
--ip <ADDRESS>	String	Exclusive Target	Target IPv4/IPv6 or private RFC-1918 address. Performs dual-mode private LAN profiling or WAN geolocation.
--username <USER>	String	Exclusive Target	Target handle. Concurrently audits account existence across 13+ digital platforms and queries breach databases.
--mac <ADDRESS>	String	Exclusive Target	Target MAC address or wireless access point BSSID. Performs IEEE OUI vendor resolution, randomized MAC detection, and spatial mapping.
--email <EMAIL>	String	Exclusive Target	Target email address. Digital footprint analysis, domain intelligence, Gravatar profiling, breach auditing & handle correlation.
--output, -o <FILE>	File Path	Optional	Exports gathered intelligence payload to a structured RFC-8259 JSON artifact file.
--concurrency, -c	Integer	Optional	Overrides maximum asynchronous worker routines (default: 15 workers).
--verbose, -v	Flag	Optional	Enables detailed debug log output and Rich tracebacks in terminal output.
--no-banner	Flag	Optional	Suppresses startup ASCII art header for clean pipeline scripting and integration.

Rapid Installation & Deployment (Linux)

```
# 1. Clone Repository
git clone https://github.com/Hacker-World-boop/OmniSight-OSINT.git
cd OmniSight-OSINT

# 2. Initialize Virtual Environment (Linux)
python3 -m venv venv
source venv/bin/activate

# 3. Install Dependencies & Make Executable
pip install -r requirements.txt
chmod +x open_osint.py

# 4. Launch Interactive Menu (Recommended)
./open_osint.py

# 5. Direct CLI Automation Mode
./open_osint.py --phone +14155552671
./open_osint.py --ip 8.8.8.8
./open_osint.py --username torvalds --output torvalds_recon.json
```

AUTHOR ATTRIBUTION & PROJECT GOVERNANCE:
OmniSight-OSINT is engineered as a modular cyber reconnaissance and threat intelligence framework by Inaam Kaleem (author = inaam kaleem) for Linux environments. Licensed under the terms of the MIT Open-Source License.